

DECK 11 · WEEK 6 · MORNING 1

Equity I Markets, Indices & Security Types

How equity trades, how we measure the market, and what we are actually buying

CFA Level I · Vol 5 Learning Modules 1–4

Roadmap — Equity I — Markets, Indices, Efficiency & Security Types

4 sections · ~2 hours · Built bottom-up from CFA readings

01

Market structure, order types & leverage

three functions of the financial system · market vs limit vs stop · limit order book · quote- vs order-driven vs brokered · call vs continuous · leverage ratio & margin call price

02

Security market indexes

the five construction & management decisions · price / equal / cap / float / fundamental weighting · divisor mechanics on a split · rebalance vs reconstitute

03

Market efficiency

intrinsic vs market value · weak / semi-strong / strong · anomalies (calendar, size, value, momentum, earnings surprise) · behavioral finance

04

Equity security taxonomy

common vs preference · callable / putable / cumulative / participating / convertible · public vs private · ADR / GDR / GRS

Learning objectives

By the end of this deck you should be able to:

1. Distinguish market, limit and stop orders, and trace how a marketable limit buy walks a live limit order book under the discriminatory pricing rule.
2. Compute the leverage ratio and the rate of return on a margin transaction, and solve for the price at which a margin call is triggered.
3. Compute the value and return of price-, equal-, market-cap- and float-adjusted indexes, and rescale the divisor correctly after a stock split.
4. State the three forms of the efficient market hypothesis, say which information set each subsumes, and give the standard counter-explanation for each documented anomaly.
5. Classify an equity security by common vs preference and by bolt-on feature, and compare access routes to non-domestic equity (direct, ADR I–III / 144A, GDR, GRS).

SECTION 01

Market structure, order types & leverage

The financial system has three jobs — serve six user purposes, discover the clearing rate of return, and allocate capital to its best use

Exchanges, OTC venues and brokered markets all exist to cut the cost of the same bilateral search: matching someone with cash to someone with an asset.

1 · SIX PURPOSES

Facilitate the six things people use the system for: (1) save for the future, (2) borrow for current use, (3) raise equity capital, (4) manage risk, (5) exchange assets for immediate or future delivery, (6) trade on information.

2 · RATE DISCOVERY

Discover the rate of return that equates aggregate savings with aggregate borrowing.

That equilibrium rate is the price of time. Every discount rate you will use later is a quote off this market.

3 · ALLOCATION

Allocate capital to its best uses — the projects with the highest risk-adjusted prospects.

This is capital-allocation efficiency: the social payoff, and the reason regulators care about market quality at all.

A WELL-FUNCTIONING SYSTEM

Complete = the instruments people need exist. Operationally efficient = commissions, spreads and price impact are low. Informationally efficient = prices track fundamental values.

Source: CFA Institute, Curriculum Vol 5, Learning Module 1, §2 "The Functions of the Financial System", pp. 4–12; well-functioning systems, pp. 60–61.

Every order carries three instruction sets — execution, validity, clearing — and market vs limit is the single most important execution choice

Market order = speed at the cost of price certainty. Limit order = price certainty at the cost of execution certainty. You never get both.

Instruction set	What it specifies	Common variants
Execution	HOW to fill — price/size conditions	Market • Limit • Marketable • AON • Hidden
Validity	WHEN the order may be filled	Day • GTC • IOC • On-close • Stop
Clearing	HOW the trade is finally settled	Long vs short • Give-up • Custodian

MARKET ORDER — TAKES THE MARKET

Executes immediately against the best resting offer. Execution is certain; price is not. In a thin book a large market order walks several price levels up and pays for every one of them.

Use when the information is perishable and delay costs more than slippage.

LIMIT ORDER — MAKES THE MARKET

Sets the worst acceptable price. If unmatched it joins the limit order book and becomes a standing offer to trade — it MAKES a market.

A buy limit priced above the best offer is "marketable": it behaves like a market order but with a bounded worst price.

Source: CFA Institute, Curriculum Vol 5, Learning Module 1, §12–13 "Orders and Execution Instructions" / "Validity and Clearing Instructions", pp. 44–51.

A limit order book ranks resting orders by price then time; "the market" is the best bid and best offer, and anything inside them makes a new market

The market is 26 bid, offered at 28. There are only four labels to learn: at the market, inside the market, behind (away from) the market, far from the market.

Bid size	Price	Offer size
	33 far	100
	31 behind	40
	29 behind	15
	28 BEST OFFER	10
—	27 inside	—
12	26 BEST BID	
25	25 behind	
40	23 behind	
100	21 far	

AT / INSIDE / BEHIND / FAR

At the market = the best bid or the best offer. Inside the market = between them; a new limit here makes a NEW market. Behind (away) and far = beyond the best, may never fill.

AGGRESSIVE VS PASSIVE

A buy at 27 is aggressively priced — it makes a new market. A buy at 29 is marketable: it crosses the best offer and trades at once. A buy at 23 is passive and waits.

ORDER PRECEDENCE HIERARCHY

- 1 Price priority — highest buy, lowest sell first.
- 2 Display precedence — shown size beats hidden size at the same price.
- 3 Time precedence — first in wins among equals.

¹ Condensed from Exhibit 2, which runs across 13 price levels (33 down to 21). Sizes are illustrative — the CFA exhibit shows prices and labels only.

Source: CFA Institute, Curriculum Vol 5, Learning Module 1, Exhibit 2 "Terms Traders Use to Describe Standing Limit Orders" + Example 21, pp. 46–47; order precedence hierarchy from §15 "Order Matching Rules", p. 57.

A 15-lot marketable limit buy clears the book from the best offer upward, then rests – CFA Example 27, walked step by step

The book on the Tokyo Stock Exchange BEFORE our order arrives. Tsubasa submits: BUY 15 contracts, limit ¥100.5, day order.

Bid size	Price (¥)	Offer size
	100.6	12 Miku
	100.5	6 Sakura
	100.4	4 Hina
10 Shou	100.3	
8 Hiroto	100.2	
15 Takumi	100.1	

SOLUTION – WALK THE BOOK UPWARD

Continuous trading, so the discriminatory pricing rule applies: each standing order fills at ITS OWN limit price.

1. Take Hina at 100.4 for 4 contracts (most aggressive offer). 11 left.
2. Take Sakura at 100.5 for 6 contracts (still at the limit). 5 left.
3. Miku is offered at 100.6 > limit 100.5 → stop. The residual 5 rest on the bid at 100.5.
4. Average price = $(4 \times 100.4 + 6 \times 100.5) / 10 = ¥100.46$.
5. A day order rests. An immediate-or-cancel order would simply have cancelled the remaining 5.

⇒ 10 filled at ¥100.46; 5 rest. Market moves 100.3/100.4 → 100.5/100.6 – same 0.10 spread.

Source: CFA Institute, Curriculum Vol 5, Learning Module 1, Example 27 "Filling a Large Order in a Continuous Trading Market", p. 58.

A limit price says WHAT price is acceptable; a stop price says WHEN the order becomes valid — they are orthogonal and can be combined

A stop-loss order does not stop losses at the stop price. It only releases an order once the market has traded through the trigger.

Order type	Valid when	Fill rule once valid	Typical use
Market	always	take best available	Immediate exit
Limit (buy)	always	fill at or below limit	Price-sensitive entry
Stop-market (sell)	trades at/below stop	take best available	Cap downside on a long
Stop-limit (sell)	trades at/below stop	fill at or above limit	Avoid bad prints
Marketable limit (buy)	always	immediate fill up to limit	Fast fill, bounded slip

THE CURRICULUM'S OWN TWO EXAMPLES — LEARN THEM AS A PAIR

Stop-loss: long at 40, submits "GTC, stop 30, market sell". If the stock prints 30 the market sell is released — and may execute well below 30 in a fast market. Unwilling to sell below 25? Then "GTC, stop 30, limit 25 sell".

Stop-buy: the manager thinks a 50 RMB stock is worth 100 but wants market confirmation first, so submits "GTC, stop 65 RMB, limit 100 RMB buy". Note the CFA caveat — leaning on the market instead of research risks Standard V.A.2.

¹ Stop-sells become valid when prices fall and stop-buys when prices rise, so stop orders add to market momentum and often execute at poor prices.

² Only a purchased put actually guarantees a selling price. The option premium is the price of that guarantee; a stop order is not insurance.

Source: CFA Institute, Curriculum Vol 5, Learning Module 1, §13 "Stop Orders", Example 23, pp. 49–51.

Secondary markets come in three execution mechanisms — quote-driven, order-driven and brokered — and most real venues blend them

Match the venue to the instrument: dealers where inventory capital is needed, exchanges where instruments are standardised, brokers where each item is unique.

Mechanism	How a trade happens	Dominant asset classes	Transparency
Quote-driven (dealer)	Trade at the dealer's quoted price	Bonds, FX, spot commodities	Low-mid
Order-driven (exchange)	Rules match buy to sell orders	Listed equities and futures	High
Brokered	Broker runs a bilateral search	Real estate, IP, big blocks	Low

WHY BROKERED MARKETS EXIST AT ALL

Dealers will not warehouse assets they cannot hedge or value — very large share blocks, real estate, fine art, intellectual property, operating companies, taxi medallions. Too few traders would ever post orders for an order-driven venue to work. So the broker sells the one thing that is scarce: a network of people who might trade.

THE THREE ARE POLES, NOT BINS

Exchanges run order-driven engines with designated dealers on top. OTC swap markets post dealer quotes on screens. Upstairs desks cross blocks away from the exchange. The CFA itself notes many trading systems use more than one structure.

Source: CFA Institute, Curriculum Vol 5, Learning Module 1, §15 "Secondary Security Market and Contract Market Structures", Example 28, pp. 55–59.

Call auctions concentrate all liquidity into one crossing price; continuous trading buys immediacy and pays for it with the spread

Many continuous markets open with a call auction and some close with one; traders who only want the closing price submit market- or limit-on-close orders.

CALL MARKET

Trades can only be arranged when the market is called. All interest is present at once, so the call itself can be extremely liquid — and between calls the market is completely illiquid.

Most calls use a single-price auction: the price that maximises total volume traded, where the order books cross.

CONTINUOUS TRADING MARKET

Trades can be arranged any time the market is open — but only if a willing counterparty (or her order) is there at that moment.

Immediacy is the product being sold. The bid–ask spread is what liquidity providers charge for supplying it and for the adverse selection they absorb.

THREE TRADE PRICING RULES — KNOW ALL THREE, NOT TWO

Uniform (call markets): every matched trade prints at the same volume-maximising price. · Discriminatory (continuous markets): the standing order's limit price sets each print, so a large order walks the book. · Derivative (crossing networks such as POSIT): the price is imported from another market, usually the midpoint of the primary exchange's best bid and offer.

Source: CFA Institute, Curriculum Vol 5, Learning Module 1, §15 "Trading Sessions" and "Execution Mechanisms", Example 26, pp. 55–59.

Leverage turns a 20% price fall into a margin call and a 50% loss of equity – the leverage ratio tells you exactly how fast

Initial margin is the minimum equity fraction at purchase. Maintenance margin — usually 25% — is the minimum you must keep. Between them sits the margin call price.

$$\begin{aligned}\text{Leverage ratio} &= \text{Position value} / \text{Equity} = 1 / \text{initial margin} \\ \text{Margin call price } P^* &= P_0 (1 - m_{\text{init}}) / (1 - m_{\text{maint}})\end{aligned}$$

where: P_0 = purchase price · m_{init} = initial margin requirement · m_{maint} = maintenance margin requirement. Equity per share after a move to P is $P_0 \cdot m_{\text{init}} + (P - P_0)$.

CFA EXAMPLE 20 — AT WHAT PRICE DOES THE CALL COME?

A trader buys stock at \$20 posting 40% initial margin. The maintenance margin requirement is 25%. Below what price does a margin call occur?

1. Equity = $40\% \times \$20 = \$8/\text{share}$; margin loan = $\$12/\text{share}$. Leverage ratio = $20 / 8 = 2.5\times = 1 / 0.40$.
2. Equity per share at any price P is $\$8 + (P - 20)$. A call comes when equity / price falls below 25%.
3. Solve $(\$8 + P - 20) / P = 0.25 \rightarrow 0.75P = 12 \rightarrow P = \16 .
4. Shortcut: $P^* = 20 \times (1 - 0.40) / (1 - 0.25) = 20 \times 0.60 / 0.75 = \16 .

⇒ Call below \$16. A 20% price fall costs 50% of the equity – 2.5× the price move, as advertised.

¹ CFA Example 19: the same stock bought at 2.5× leverage falls from \$20 to \$15 and the equity return is –68.9%, not –25%.

² Short sellers post margin too: their equity is eroded when prices RISE, and the broker buys the stock back if the account is not topped up.

Source: CFA Institute, Curriculum Vol 5, Learning Module 1, §11 "Leveraged Positions", Examples 19–20, pp. 41–44.

SECTION 02

Security market indexes

An index is five decisions — target market, selection, weighting, rebalancing, reconstitution — and every one of them shapes the reported return

An index is an opinion about what "the market" means, written down as rules. Two indexes over the same universe can report very different returns on the same day.

Decision	What it fixes	Illustration
1 · Target market	Asset class, geography, exchange, sector, size, style	S&P 500: US large cap
2 · Security selection	All constituents or a sample; fixed N or variable N	FTSE 100 fixes N
3 · Weighting	Price / equal / market-cap / float-adjusted / fundamental	Dow: price-weighted
4 · Rebalancing	How often weights reset to the rule (usually quarterly)	Equal-W: high turnover
5 · Reconstitution	When selection and weighting rules are re-applied	Russell 2000: annual

RULES-BASED VS COMMITTEE — AND WHY IT MATTERS

Objective, mechanical rules determine the constituents of most indexes: the TOPIX takes every stock meeting the Tokyo First Section criteria, and a stock that stops meeting them drops out automatically. But not all indexes work that way — the CFA names the S&P 500 and the S&P BSE SENSEX as using a selection committee and more subjective decision rules.

Rules-based means predictable and backtestable; committee-run means human judgement can handle mergers, distress and sector balance — at the cost of some opacity for anyone trying to front-run the change.

Source: CFA Institute, Curriculum Vol 5, Learning Module 2, §3 "Index Construction" (the five questions, p. 86) and §4 "Index Management", pp. 86–96.

An index return is a weighted average of constituent returns — price and total return differ only by the income leg, taken at index level

Beginning-of-period weights determine each constituent's contribution. End-of-period weights only tell you where the index has drifted to.

$$\begin{aligned} \text{PR}_I &= \sum w_i \text{PR}_i & \text{TR}_I &= \sum w_i \text{TR}_i \\ \text{TR}_i &= (\text{P1}_i - \text{P0}_i + \text{Inc}_i) / \text{P0}_i \end{aligned}$$

where: w_i = BOP weight of constituent i · P0 , P1 = price at start and end of the period · Inc_i = income per unit. At index level, income is divided by the divisor first.

MINI-EXAMPLE — TWO-STOCK PRICE-WEIGHTED INDEX, DIVISOR 2

HELION: 50.00 → 55.00, dividend 0.75. RENKO: 25.00 → 22.00, dividend 0.10. Divisor = 2 (= number of constituents at inception).

1. Index value: BOP = $(50 + 25) / 2 = 37.50$. EOP = $(55 + 22) / 2 = 38.50$.
2. Price return = $(38.50 - 37.50) / 37.50 = +2.67\% = 0.6667(+10.00\%) + 0.3333(-12.00\%)$.
3. Income at INDEX level = $(0.75 + 0.10) / 2 = 0.425$ points — divide by the divisor first.
4. Total return = $(38.50 - 37.50 + 0.425) / 37.50 = 1.425 / 37.50 = +3.80\%$.

⇒ Price return 2.67% vs total return 3.80% — the income leg is worth 113 bps this period.

Price weighting lets the highest-quoted share dominate by accident of quotation — nothing to do with the size of the company

A \$200 share carries ten times the weight of a \$20 share regardless of which company is bigger. This is the Dow Jones Industrial Average's original sin.

$$W_i = P_i / \sum P_j$$

$$\text{Index value} = \sum P_i / \text{Divisor}$$

where: The divisor is set at inception, typically to the number of constituents, then rescaled on splits and constituent changes so the level is preserved.

Security	BOP price	BOP weight	EOP price	Price return	EOP weight
HELION	50.00	49.26%	55.00	+10.00%	52.38%
RENKO	25.00	24.63%	22.00	-12.00%	20.95%
AZURIS	12.50	12.32%	8.00	-36.00%	7.62%
NOVEX	10.00	9.85%	14.00	+40.00%	13.33%
CORYN	4.00	3.94%	6.00	+50.00%	5.72%
Index (D = 5)	101.50	100.00%	105.00	+3.45%	100.00%

THE HEADLINE

Index 20.30 → 21.00, price return +3.45%, total return +4.33%. HELION carries 49.26% of the weight on 26.29% of the group's market cap.

Source: CFA Institute, Curriculum Vol 5, Learning Module 2, §3 "Price Weighting", Exhibit 1, pp. 87–88; market caps from Exhibit 4, p. 92.

A stock split changes every weight in a price-weighted index and none in a cap-weighted one — so the divisor, not the level, must move

At end of period HELION splits 2-for-1: price 55.00 → 27.50, shares 3,000 → 6,000. Market capitalisation is unchanged. Yet Σ prices drops from 105.00 to 77.50.

WORKED EXAMPLE — RESCALING THE DIVISOR

1. Before the split: Σ prices = 55 + 22 + 8 + 14 + 6 = 105.00; divisor 5.00 → index = 21.00.
2. After the split: Σ prices = 27.50 + 22 + 8 + 14 + 6 = 77.50 — but nothing economic has changed.
3. New divisor = Σ prices after ÷ index before = 77.50 / 21.00 = 3.69, so the index still reads 21.00.
4. Cap-weighted: HELION 3,000 → 6,000 shares at 27.50 → cap still 165,000; total still 579,000.

⇒ Price-weighted divisor 5.00 → 3.69. Cap-weighted: the split is a complete non-event.

Security	Price post-split	P-W weight after	Mkt cap (EOP)	Cap-W weight
HELION	27.50	35.48%	165,000	28.50%
RENKO	22.00	28.39%	220,000	38.00%
AZURIS	8.00	10.32%	40,000	6.91%
NOVEX	14.00	18.07%	112,000	19.34%
CORYN	6.00	7.74%	42,000	7.25%

¹ Shares outstanding are the curriculum's own: 3,000 / 10,000 / 5,000 / 8,000 / 7,000 for HELION, RENKO, AZURIS, NOVEX and CORYN (Exhibit 4, p. 92).

Source: CFA Institute, Curriculum Vol 5, Learning Module 2, Exhibit 2 "Impact of 2-for-1 Split in Security A", p. 89; cap weights from Exhibit 4, p. 92.

Each weighting method solves one problem and creates another — and the bias it creates is the exam question, every time

Float-adjusted market-cap weighting is now the industry default — the CFA notes that unless stated otherwise, "market-capitalization weighting" means float-adjusted.

Method	Weight rule	Advantage	Disadvantage / effect
Price-weighted	$w_i = P_i / \sum P_j$	Simplicity	Splits reshuffle all weights
Equal-weighted	$w_i = 1/N$ at inception	Simplicity	Big names under-represented
Market-cap	$w_i = Q_i P_i / \sum Q_j P_j$	Held at market value	Overweights what has risen
Float-adjusted cap	$w_i = f_i Q_i P_i / \sum f_j Q_j P_j$	Only tradable shares	Momentum tilt, as with cap
Fundamental	$w_i = F_i / \sum F_j$	Price-independent	Deliberate value tilt

GET THE BIAS ON THE RIGHT SCHEME

Equal weighting: two disadvantages, not one — the largest names are under-represented AND the weights stop being equal the moment prices move, so it needs frequent rebalancing.

Market-cap weighting: the CFA calls the effect similar to a MOMENTUM strategy — whatever has risen (and may be overvalued) gets a bigger weight, and float adjustment does NOT remove that tilt.

Fundamental weighting (F = book value, cash flow, revenue, earnings, dividends, employees): a value tilt by construction, and a CONTRARIAN effect on rebalancing. Price weighting is never rebalanced; cap weighting largely rebalances itself.

¹ CFA p. 93 treats float-adjusted weighting as market-cap weighting for advantages and disadvantages: float changes the share count, not the tilt.

Source: CFA Institute, Curriculum Vol 5, Learning Module 2, §3 "Index Weighting", price / equal / cap / float-adjusted / fundamental, pp. 87–95.

Rebalancing refreshes the weights, reconstitution refreshes the membership — two different kinds of turnover a tracker must absorb

Price-weighted indexes are never rebalanced; cap-weighted indexes largely rebalance themselves; equal-weighted indexes must be rebalanced constantly. All of them get reconstituted.

REBALANCING — WEIGHTS

Adjust the constituents' weights back to the rule, on scheduled rebalancing dates, usually quarterly.

In the CFA's equal-weighted example the weights had already drifted from 20% each to 19.93 / 15.94 / 11.60 / 25.36 / 27.17 in a single period — so the winners get sold and the losers bought.

RECONSTITUTION — MEMBERSHIP

On the reconstitution date the provider re-applies the inclusion criteria, drops what no longer qualifies, adds what does — then re-applies the weighting method to everyone.

So for a cap-weighted index one deletion forces a weight change across ALL remaining names, not just the two that moved.

WHY RECONSTITUTION IS NOT A BACK-OFFICE EVENT — THE CURRICULUM'S OWN CASE

The Russell 2000 reconstitutes annually, and managers front-run it: they buy what they think will be added and sell what they think will be dropped. Uranium Energy Corporation was bid up from late April 2009; by the 12 June preliminary announcement it had already risen more than 300%, and it closed the 30 June reconstitution date up nearly 400% for the quarter. That is the index-inclusion effect, paid for by the tracker.

Source: CFA Institute, Curriculum Vol 5, Learning Module 2, §4 "Index Management: Rebalancing and Reconstitution", Exhibit 6, pp. 95–97.

Equity indexes come in four families — broad market, multi-market, sector and style — and each answers a different analytical question

Pick the index that matches the question. "Did US large caps rise?" → broad. "Did tech beat staples?" → sector. "Did value beat growth?" → style.

Family	What it represents	Typical use	Examples
Broad market	A whole domestic equity market	Sentiment, benchmarking	S&P 500, TOPIX
Multi-market	Several countries or regions	Global / EM vs DM tilt	MSCI ACWI, MSCI EAFE
Sector	One economic sector	Sector rotation analysis	MSCI World Energy
Style	Size × value / growth	Style-drift, factor tilt	Russell 1000 Growth

THE SIX STYLE CATEGORIES

Three size buckets (large / mid / small) × two styles (value / growth) = six categories: large-cap value, large-cap growth, mid-cap value, mid-cap growth, small-cap value, small-cap growth.

Stocks migrate between boxes on reconstitution, so style indexes turn over far more than broad market indexes.

THE FIVE USES OF AN INDEX

- 1 Gauge of market sentiment.
- 2 Proxy for measuring and modelling return, systematic risk and risk-adjusted performance.
- 3 Proxy for an asset class in allocation models.
- 4 Benchmark for actively managed portfolios.
- 5 Model portfolio for index funds and ETFs.

Source: CFA Institute, Curriculum Vol 5, Learning Module 2, §6 "Equity indexes", pp. 99–102; uses of market indexes, §5, pp. 97–99.

SECTION 03

Market efficiency

Efficiency is a continuum, not a switch — and a market is inefficient only if you can beat it AFTER costs

Intrinsic value must be estimated; market value is observed. Active management is the bet that the gap between them is real, exploitable and large enough to pay for itself.

$$\text{Abnormal return} = \text{Actual return} - \text{Expected return}$$

where: Expected return means risk-adjusted. Consistently earning abnormal returns by trading on a given information set is the evidence AGAINST efficiency with respect to that set.

Pushes a market TOWARD efficiency

More participants: investors and analysts

Fair-disclosure rules (e.g. SEC Regulation FD)

Active arbitrage and low operational frictions

Opening the market to foreign investors

Pushes a market AWAY from efficiency

High information-acquisition costs

Transaction costs and wide spreads

Short-selling limits and borrow costs

Thin trading and illiquid instruments

THE MODERN VIEW — EFFICIENT WITHIN THE BOUNDS OF TRANSACTION COSTS

Two identical assets can trade at different prices and both prices are efficient if the gap is smaller than the cheapest trader's cost of arbitraging it. Grossman and Stiglitz add the other half: prices must offer SOME return to information acquisition, or nobody would gather information at all.

Source: CFA Institute, Curriculum Vol 5, Learning Module 3, §2 "The Concept of Market Efficiency" and §3 "Factors Affecting Market Efficiency", pp. 125–132; abnormal return defined p. 133.

The three forms nest — weak inside semi-strong inside strong — and developed-market evidence supports the first two but rejects the third

Each form is a claim about which information is ALREADY in the price — and therefore about which analysis cannot earn abnormal returns.

Form	Prices reflect	Strategy that cannot work	Developed-market evidence
Weak	Past prices and volume	Technical analysis	Largely supported
Semi-strong	Weak + all public info	Fundamental analysis	Broadly supported
Strong	Public + private info	Trading on inside information	Rejected by evidence

WHY THEY NEST

Public information includes past market data, so semi-strong efficiency implies weak-form efficiency. Private plus public includes both, so strong-form implies the other two. The evidence runs the other way: markets clear the first hurdle, mostly clear the second, and fail the third because insiders demonstrably earn abnormal profits.

IMPLICATIONS — AND THE PARADOX

Technical analysts keep the market weak-form efficient by arbitraging away the patterns they find; fundamental analysts keep it semi-strong efficient by disseminating value-relevant information. Both destroy the edge they are hunting. Mutual funds perform roughly like the market before fees and worse after them — the case for passive management.

Source: CFA Institute, Curriculum Vol 5, Learning Module 3, §4 "Forms of Market Efficiency", pp. 132–137; implications, §5, pp. 137–138.

Anomalies fall into three research categories — time-series, cross-sectional and other — and most dissolve once risk-adjusted and costed

The three-part test for a real anomaly: does it replicate out of sample, does it survive risk adjustment, and is it still profitable after transaction costs?

Anomaly	Category	Observation	Standard counter-explanation
January effect	Time series	Small caps spike in early January	Gone once risk-adjusted
Momentum	Time series	Short-term winners keep winning	Rational cash-flow shocks
Size effect	Cross-sectional	Small caps beat large, risk-adjusted	Microcaps only (FF 2008)
Value effect	Cross-sectional	Low P/E and P/B beat growth stocks	FF 3-factor: it is risk
Earnings surprise	Other	Prices drift AFTER the announcement	Costs and risk not netted
Closed-end discount	Other	Funds trade 4–10% below NAV	Arbitrage costs kill it
IPO long-run returns	Other	Day-1 pop, then weak long-run	Small-sample methodology

EXHIBIT 4 — FOUR MORE CALENDAR EFFECTS, ALL LARGELY GONE

Turn-of-the-month: returns higher on the last trading day of the month and the first three of the next. Day-of-the-week: Monday negative, the other four positive. Weekend: weekend returns below weekday returns. Holiday: returns higher the day before a market holiday. Most have been eliminated over time — either arbitrated away, or never really there.

¹ Fama and French (1995): apply the three-factor model instead of the CAPM and the value anomaly disappears — it was a risk factor all along.

² Fama (1998): fix the methodology and most anomalies vanish; over-reactions are as common as under-reactions, so markets are efficient on average.

Source: CFA Institute, Curriculum Vol 5, Learning Module 3, §6–7 "Market Pricing Anomalies", Exhibits 3–4, pp. 139–145.

Behavioral biases give anomalies a mechanism — but efficiency only requires that the **MARKET** be rational, not that every investor is

If some investors deviate from rationality, others are assumed to observe the deviation and trade against it. Those responses are what move the market back toward efficiency.

LOSS AVERSION

Investors dislike losses more than they like equivalent gains — a stronger preference than plain risk aversion.

If loss aversion dominates, we should observe OVERREACTION in prices.

The counter (Fama 1998): under-reaction turns out to be about as common as over-reaction, so the bias alone cannot explain the evidence.

HERDING & CASCADES

Herding: investors trade the same side of the same securities at once, ignoring their own information. It can produce EITHER under- or over-reaction, depending on which way the herd runs.

An information cascade is the related but distinct idea: those who act first transmit information, and later traders imitate them.

Cascades can be rational — if the informed act first, imitation helps prices incorporate information.

OVERCONFIDENCE

Investors overestimate their ability to process and interpret information, so they misprice securities.

Evidence of overconfidence mispricing is reported in the US, UK, German, French and Japanese markets — concentrated in higher-growth companies, whose prices react slowly.

Most researchers argue the mispricing is temporary. It is exploitable only if the correction is slow AND predictable.

SECTION 04

Equity security taxonomy

Common shares are the residual claim with the votes; preference shares rank ahead on cash flow and liquidation but usually have no vote at all

Preference shares are a hybrid: bond-like fixed dividends, equity-like perpetual life — and behind debt, ahead of common, in liquidation.

Feature	Common shares	Preference shares
Dividend	Discretionary; can be zero	Fixed rate, not contractually owed
Dividend priority	Last, after preferred and debt	Ahead of common; accrues if cumul.
Liquidation priority	Last (residual claimant)	Ahead of common, behind debt
Voting rights	One vote per share (typical)	None unless granted at issuance
Share in earnings growth	Full — price and dividend growth	Capped unless participating
Maturity	Perpetual	Usually perpetual; may be called

STATUTORY VS CUMULATIVE VOTING — THE CURRICULUM'S OWN ARITHMETIC

Statutory voting: one vote per share, per candidate. Own 100 shares with four directors to elect and you may cast at most 100 votes for each.

Cumulative voting: total votes = shares × directors being elected = $100 \times 4 = 400$, and you may pile all 400 onto a single candidate. That is the whole point — it lets a small holder win one board seat instead of losing every seat by the same margin.

Source: CFA Institute, Curriculum Vol 5, Learning Module 4, §2 "Characteristics of Equity Securities" — Common Shares and Preference Shares, pp. 165–170.

Five bolt-on features turn a plain preference share into five economically different securities – each one is an embedded option

Each feature moves value between issuer and investor. The holder's return is the stated dividend plus the net value of every embedded option.

Feature	Option held by	What it does	Value moves to
Callable	Issuer	Issuer buys back at a set price	Favours issuer
Putable	Investor	Investor sells back at a set price	Favours investor
Cumulative	Investor (automatic)	Unpaid dividends accrue and must be paid	Favours investor
Participating	Investor (automatic)	Extra dividend if profits beat a threshold	Favours investor
Convertible	Investor	Swap into a set number of common shares	Favours investor

THE CURRICULUM'S FOUR ADVANTAGES OF CONVERTIBLE PREFERENCE SHARES

1 A higher dividend than the company's common shares. 2 The opportunity to share in the company's profits. 3 The chance to benefit from a rise in the common share price, through the conversion option. 4 Less price volatility than the common, because the dividend is known and stable.

That asymmetry is exactly why convertible preference shares are a popular financing route in venture capital and private equity, where the issuer is high-risk and an IPO may be years away. Note also: participating preference shares are most common in smaller, riskier companies, where liquidation is a live concern.

Source: CFA Institute, Curriculum Vol 5, Learning Module 4, §2 "Preference Shares", Exhibits 9–10 (GDL Fund, TEN Ltd), pp. 168–170.

A convertible preference share decomposes into a fixed-income floor plus an embedded call on the common – floor below, equity above

Bond floor = value as a straight preference share. Conversion value = conversion ratio $\times$ common price. The traded price sits above the higher of the two.

$$\text{Price} = \max(\text{Bond floor}, \text{Conversion value}) + \text{Time value}$$

where: Bond floor = PV of the preference dividend stream at the yield on comparable straight preferreds · Conversion value = conversion ratio $\times$ common share price.

ILLUSTRATION (NOT A CFA EXAMPLE) – AZURIS 6% CONV. PREF.

Par 50 · dividend 6% of par = 3.00 a year, perpetual · conversion ratio 2.5 · AZURIS common at 18.00 · comparable straight preferreds yield 7.14% · convertible trades at 48.00.

1. Bond floor = perpetuity $3.00 / 0.0714 = 42.00$. This is the downside if the equity story fails.
2. Conversion value = $2.5 \times 18.00 = 45.00$. This is what the shares are worth if converted today.
3. Intrinsic value = $\max(42.00, 45.00) = 45.00$ — the conversion option is in the money.
4. Time value = traded 48.00 – intrinsic 45.00 = 3.00, the value of waiting rather than converting.

⇒ 48.00 = 45.00 intrinsic + 3.00 time value; 6.00 above the 42.00 floor. Bond below, equity above.

¹ The curriculum does not put numbers on this decomposition; the maths is standard convertible arithmetic and returns in the fixed-income deck.

² Conversion ratio is fixed at issuance, so conversion value moves one-for-one with the common price. The floor moves with credit spreads and rates.

Source: CFA Institute, Curriculum Vol 5, Learning Module 4, §2 "Preference Shares", pp. 169–170. The arithmetic on this slide is an instructor illustration, not a CFA example.

Private equity comes in three types – venture capital, leveraged buyout and PIPE – with no quoted price, no active secondary market and no mandated disclosure

Private equity securities are placed with institutions in non-public offerings, so they have no market-determined price, are highly illiquid, and require negotiation to trade at all.

VENTURE CAPITAL

Seed, early-stage or mezzanine capital for companies still building product and growth.

Exit is typically 3 to 10 years from start-up, via IPO or sale to other investors — and the exit price is the whole return.

Instrument of choice: convertible preference shares, increasingly so as holding periods lengthen.

LEVERAGED BUYOUT

A group uses a large amount of debt to buy all the shares of a listed company; the shares stop trading and the company is taken private. If the buyers are mainly existing management it is an MBO.

Targets: undervalued assets that can be sold to cut debt, plus high cash flows to service it.

Objective: restructure, then re-list.

PIPE

Private investment in public equity. A listed company needs capital fast and sells a sizeable stake to a private investor or group.

Triggers: a big expansion opportunity, heavy indebtedness, or a rapid deterioration in operations.

The more urgent the need and the larger the cheque, the deeper the discount to the quoted price.

Source: CFA Institute, Curriculum Vol 5, Learning Module 4, §3 "Private Versus Public Equity Securities", Exhibit 11, pp. 170–172.

Non-domestic equity reaches you three ways — direct, a depository receipt, or a global registered share — and the ADR levels differ on one thing: capital

A depository receipt is created when a bank holds the foreign shares and issues receipts against them, in the local currency, at a stated ratio — so a cross-border trade becomes a local one.

Route	Trades on	SEC registration	Raise capital?	Typical investor
Direct investing	Issuer's home exchange	None extra	n/a	Institutional
ADR Level I	US OTC market	Form F-6	No	Retail + inst.
ADR Level II	NYSE / NASDAQ / AMEX	Form F-6	No	Broad US market
ADR Level III	NYSE / NASDAQ / AMEX	Forms F-1 and F-6	Yes — public	Broad US market
ADR Rule 144A	Private placement, PORTAL	None	Yes — private	QIBs only
GDR	London, Luxembourg	No SEC	Yes	Non-US inst.
Global registered share	Several exchanges at once	Each market	Yes	Global inst.

SPONSORED VS UNSPONSORED — THE DISTINCTION THAT DECIDES WHO VOTES

Sponsored: the foreign issuer is directly involved in the programme, and holders get the same rights as direct shareholders, including the vote and the dividend. Un-sponsored: the depository buys the shares itself and issues receipts through brokers — the DEPOSITORY keeps the voting rights, not you. Sponsored DRs face greater reporting requirements; in the US they must be registered with the SEC. Levels I–III are all sponsored.

Source: CFA Institute, Curriculum Vol 5, Learning Module 4, §4 "Non-Domestic Equity Securities", Exhibits 13–14, pp. 173–178.

Deck 11 — Equity I — Markets, Indices, Efficiency & Security Types

- Three functions, six purposes. Every order carries execution, validity and clearing instructions; market orders TAKE liquidity, limit orders MAKE it.

- Leverage ratio = $1 / \text{initial margin}$. Margin call at $P^* = P_0(1 - m_{\text{init}})/(1 - m_{\text{maint}})$: \$20 bought on 40% margin, 25% maintenance, calls at \$16.

- Cap weighting drifts toward momentum; fundamental weighting is a value tilt and a contrarian rebalance; equal weighting under-represents big names and churns.

- Preference shares trade upside for priority. Five bolt-ons: callable (issuer), putable, cumulative, participating, convertible — the last four favour the investor.

- Books rank by price, then display, then time. Continuous markets use the discriminatory rule, so an arriving order pays each standing order's own limit price.

- An index is five decisions. On a split the price-weighted DIVISOR is rescaled ($5.00 \rightarrow 3.69$) to hold the level; the cap-weighted index does not move at all.

- Weak $\subset$ semi-strong $\subset$ strong. Developed markets support the first two and reject the third. Anomalies must survive replication, risk adjustment AND costs.

- Non-domestic access: direct, ADR I–III and Rule 144A, GDR, GRS. Only Level III and 144A raise new capital; only sponsored DRs carry the vote.